

Q1. Conversion of Cartesian coordinates to polar coordinates:

Transform the equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ into polar co-ordinates.

Q2. Evaluate $\lim_{\substack{x \rightarrow \infty \\ y \rightarrow 2}} \frac{xy + 4}{x^2 + 2y^2}$

Q3. Evaluate $\lim_{\substack{x \rightarrow 1 \\ y \rightarrow 2}} \frac{2x^2y}{x^2 + y^2 + 1}$

Q4. If $u = (x^3 + 2x^2y + y^3)^{\frac{3}{2}}$, then show that

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{63}{4} (x^3 + 2x^2y + y^3)^{\frac{3}{2}}$$

Q5. If z is a function of x and y and u and v be two other variables, such that $u = lx + my$, $v =$

$ly - mx$, show that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = (l^2 + m^2) \left(\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2} \right)$.

Q6. Find n , if for $\theta = t^n e^{\frac{r^2}{4t}}$ & $\frac{1}{r^2} \left(\frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) \right) = \frac{\partial \theta}{\partial t}$.

Q7. If $z = f(x + ay) + g(x - ay)$, then prove that $z_{yy} = a^2 z_{xx}$.

Q8. If $z = x^2 + 2y^2$, $x = r \cos \theta$, $y = r \sin \theta$, then find

$$(i) \quad \left(\frac{\partial z}{\partial x} \right)_y \quad (ii) \quad \left(\frac{\partial z}{\partial \theta} \right)_r$$

Q9. State and Prove Euler's Theorem.

Q10. Find the degree of $f(x, y) = \log y - \log x$.

Q11. If $u = r^n$ & $r^2 = x^2 + y^2 + z^2$ then Prove that $u_{xx} + u_{yy} + u_{zz} = n(n+1)r^{n-2}$

Q12. If $u = \log(x^3 + y^3 - x^2y - xy^2)$, then Prove that $u_{xx} + u_{yy} + u_{zz} = \frac{-4}{(x+y)^2}$.

Q13. $x^x y^y z^z = C$, then show that at $x=y=z$, $\frac{\partial^2 z}{\partial x \partial y} = -(x \ln ex)^{-1}$.